

Living leaves



FEATHERY LEAVES
Water plants often have feathery leaves to allow the water to flow past without damaging them.

LEAVES ARE SO VARIED that botanists have invented a whole new language to describe their shapes and the way they are fixed to plants. One reason for all this variety is that each species of plant faces its own challenges in harvesting sunlight (pp. 14–15). A plant living on the gloomy floor of a rain forest, for example, may need big leaves to catch enough light. A plant growing on top of a cliff has plenty of light, but it is lashed by strong winds, so it needs small, strong leaves if it is to survive. Some plants have more than one type of leaf. This is most marked in plants that start their lives under water but then flower above it.

One example is the water crowfoot. Its submerged leaves are fine and feathery, to let water flow past without tearing them, while its upper leaves are flat and broad, so that they float on the surface.

CHANGING COLOR
The leaves of the herb Robert change from green to crimson as fall approaches, or in very dry weather.



FURRY LEAVES
Some leaves have a furlike covering that helps to reduce water loss. These leaves are from a cultivated pyrethrum, which is grown in gardens.



PARALLEL VEINS
The leaves of plants such as grasses, orchids, and lilies have parallel veins (p. 9). These straplike leaves are from a member of the lily family.



Older leaves

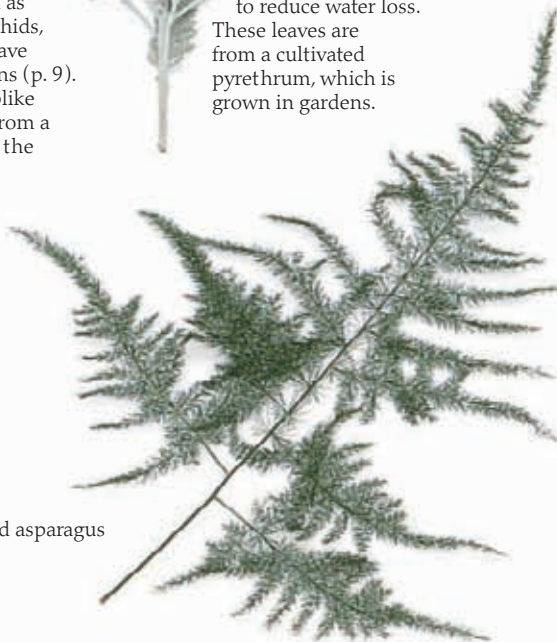
DIFFERENT SHAPES
This eucalyptus tree has leaves of two totally different shapes. The leaves of the young stems are round, like coins, and each one completely encircles the branch. The leaves on the older parts of the stems have stalks and are shaped like short straps.



Young leaves

Leaf supported by strong ribs

FACING THE WIND
Wild asparagus lives on windy coasts. Instead of true leaves, it has feathery, green, leaflike stems, called cladodes, which can withstand gales. Large, fleshy leaves would be torn to pieces.



Wild asparagus

WATERSIDE GIANTS
Gunneras grow on riverbanks in tropical forests, but are sometimes found beside water in warmer parts of temperate countries. Their leaves can be enormous—as much as 6 ft (2 m) in diameter.



Underside of a section of a gunnera leaf